RAJALAKSHMI ENGINEERING COLLEGE

P Saam 251801276-2025-AIDS

P2

Started on	Thursday, 27 August 2026, 9:01 AM
State	Finished
Completed on	Thursday, 27 August 2026, 9:01 AM
Time taken	38 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given two arrays array_One[] and array_Two[] of same size N. We need to first rearrange the arrays such that the sum of the product of pairs(1 element from each) is minimum. That is SUM (A[i] * B[i]) for all i is minimum.

For example:

Input	Result
3 1 2 3 4 5 6	28

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

// Comparison function for ascending order
int compareAscending(const void *a, const void *b) {
    return (*(int *)a - *(int *)b);
}

// Comparison function for descending order
int compareDescending(const void *a, const void *b) {
    return (*(int *)b - *(int *)a);
}

int main() {
    int n;
    // Read the size of the arrays
    if (scanf("%d", &n) != 1) return 0;

    int *array_One = (int *)malloc(n * sizeof(int));
    int *array_Two = (int *)malloc(n * sizeof(int));

    // Read elements for array_One
    for (int i = 0; i < n; i++) {
        scanf("%d", &array_One[i]);
    }

    // Read elements for array_Two
    for (int i = 0; i < n; i++) {
        scanf("%d", &array_Two[i]);
    }

    // Sort array_One in ascending order
    qsort(array_One, n, sizeof(int), compareAscending);

    // Sort array_Two in descending order
    qsort(array_Two, n, sizeof(int), compareDescending);

    long long min_product_sum = 0;

    // Calculate the sum of products of corresponding elements
    for (int i = 0; i < n; i++) {
        min_product_sum += (long long)array_One[i] * array_Two[i];
    }

    // Print the minimum sum
    printf("%lld\n", min_product_sum);

    // Free allocated memory
    free(array_One);
    free(array_Two);

    return 0;
```

	Input	Expected	Got	
✔	3 1 2 3 4 5 6	28	28	✔
✔	4 7 5 1 2 1 3 4 1	22	22	✔
✔	5 20 10 30 10 40 8 9 4 3 10	590	590	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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